GridSense AI - Real-Data Migration, Validation and Audit Report

built 2026-11-06 07:49 UTC - every figure measured at build time, none typed in

1. Summary

GridSense AI used to run on a synthetic stand-in dataset that existed to prove the pipeline mechanically before any API credentials were available. It now runs on real hourly data pulled from the ENTSO-E Transparency Platform (load) and NASA POWER (weather) for three bidding zones, with every model retrained, the uncertainty bands re-calibrated, and the whole release-gate suite re-played on the real held-out window. Making that switch found six defects that a synthetic series structurally cannot expose, and it also made one release gate go from PASS to FAIL: the real grid is harder than the simulation, and this report says so rather than moving the goalposts.

Backend	https://github.com/huzaifahimad/Grid_Sense.AI-Backend- https://grid-senseai-backend.fastapicloud.dev (root routes, no /api/v1 prefix)
Frontend	https://github.com/huzaifahimad/Grid_Sense.AI-frontend- https://grid-senseai.vercel.app (HTTP 200)
Training frame	51,959 hourly rows, 29 columns, 2024-11-11 to 2026-11-04 UTC
Served window	1,008 rows held in memory by the API (newest hour 2026-11-04T17:00), stamped to the same sources. Nothing refetches it, so the newest served hour ages one hour per hour between rebuilds.
Load source	entsoe: 51,959 rows
Weather source	nasa_power: 51,689 rows, unavailable: 270 rows
Capacity basis	per-zone p99 of observed load (proxy, not rated)
Release gates	9 PASS / 4 WARN / 0 FAIL of 13 on 3,464 replayed test hours
Test suite	59 collected - 59 passed, 1 warning in 35.99s
Commits	backend fcf0744, frontend b44f979

All 51,959 training-frame rows are stamped to a real measured source - load=['entsoe'] weather=['nasa_power', 'unavailable'], and the served window carries the same stamps. No stand-in row remains.

2. Why the data was synthetic, and what made it real

ingestion/synthetic_bootstrap.py writes files with exactly the schema the real clients produce. That is why every downstream stage ran unchanged against it, and also why the distinction was easy to lose. The API was live and correct; the values it reasoned over were invented by a seeded formula:

```
load = base_load * (1 + diurnal + weekly + seasonal + trend + noise), clipped to [0.4, 1.6] * base_load    with    np.random.seed(42)
```

The clip ceiling was detectable in the data: the synthetic Greece zone peaked at exactly 4640.0 MW = 2900 x 1.6. A real grid has no maximum that is a round multiple of its mean. Both real clients now stamp a source column, build_features refuses to merge unstamped frames, and GET /health publishes what the deployed service is actually serving:


```
{
 "load": {
  "entsoe": 1008
 },
 "weather": {
  "nasa_power": 738,
  "unavailable": 270
 },
 "capacity": "p99-of-load proxy",
 "load_latest_timestamp": "2026-11-04T17:00:00+00:00",
 "weather_latest_timestamp": "2026-10-31T23:00:00+00:00",
 "weather_lag_hours": 90.0,
 "is_synthetic": false
}
```

Six defects the real pull exposed and the synthetic one could not

#	Defect	Consequence had it shipped	Fix
1	entsoe_client.pull_load assumed a Series; the live SOP endpoint returns a DataFrame named 'Actual Load'	A blind two-column rename worked by luck for one column and broke the moment ENTSO-E added a second document type	Read iloc[:, 0] explicitly and fail loudly on unexpected shapes
2	Resolution differs per zone: ES publishes 15-minute, GR and IT_SOUTH hourly	load_lag_1h would mean '15 minutes ago' for Spain and '1 hour ago' for Greece, because every lag feature is indexed by row position	Resample to hourly means after ingest; one grid for all zones
3	The SOP index is the zone's civil timezone with DST (+01:00/+02:00), not UTC	Merging a civil-time index against UTC weather silently shifted the entire diurnal cycle by 1-2 h	tz_convert('UTC') before resampling - civil->UTC is injective, so the repeated autumn hour stays two distinct UTC hours
4	nasa_power_client returned naive timestamps and a -999 missing sentinel	A naive key merged against a tz-aware key raises, not warns; -999 left in place would have trained on invented weather	tz_localize('UTC') and NaN the sentinel. Verified empirically: mean temperature peaks at 14:00 UTC for Madrid, 13:00 for Athens, 12:00 for Naples - after solar noon in UTC, not local time
5	Hours the TSO had not yet republished were dropped permanently, and identically across all three zones (58 ES / 23 GR / 23 IT_SOUTH)	Real data silently lost: identical missing UTC hours across three independent TSOs are an ingestion artifact, not a grid event, and each hole corrupts the lag block around it	Re-pull proved the API now returns those hours; pull_load re-requests every empty day (bounded by MAX_BACKFILL_DAYS) and fills only hours that were empty
6	The newest 72 h of load arrived with no weather at all (NASA POWER publishes roughly two days behind)	LightGBM never raises on NaN - it routes missing values down the learned default direction and returns a confident number whose accuracy has never been measured. The live forecast would have been blind on exactly the most recent rows	build_features now requires complete weather for a row to be usable, so the serving window ends where both sources exist; covered by a regression test

3. Row-count vs time-correctness: the class of bug synthetic data hides

groupby().shift(N) counts rows, not hours, and the conformal calibrator sliced history by row offset too. With a complete synthetic grid those are the same thing, so both were invisible until real publication gaps appeared. Two fixes, each with a test that fails without it:

- features/build_features.reindex_hourly_grid() inserts an empty row for every hour a TSO did not publish, so the row after a hole carries NaN instead of a stale value labelled as fresh. The test builds a 400 h frame with a 3-hour hole and asserts the naive shift labels a 4-hour-old value as load_lag_1h, while the corrected frame leaves load intact and NaNs the lag.
- models/calibrate_forecast_intervals.build_origins() rejects any origin whose 168 h history plus 48 h horizon straddles a gap - shared by the calibrator and the closed-loop validator, so one fix protects both. On this pull it removed 1,573 candidate origins per zone in the calibration window, and 2 gap hours remain in the test window ({'ES': 2, 'GR': 0, 'IT_SOUTH': 0}).

4. Real data now in the model

Zone	Rows	Holes	Span (UTC)	Load min	Load max	Capacity proxy
ES	17,257	94	2024-11-11 to 2026-11-04	17,643	41,104	37,215 MW
GR	17,351	0	2024-11-11 to 2026-11-04	3,207	10,370	8,875 MW
IT_SOUTH	17,351	0	2024-11-11 to 2026-11-04	683	4,563	3,735 MW

The magnitudes are their own sanity check: Spain 20-35 GW, Greece 4-9 GW, southern Italy 1.4-4 GW, with a visible winter/summer swing and Madrid temperatures from -4.9 to 40.6 degC. The synthetic generator was pinned to a 5,200 MW base and could not have produced Spain at 35 GW.

5. Are the predictions actually computed?

Is any served number a stored constant, or does it come from the committed model applied to the committed data? scripts/verify_prediction_integrity.py answers it by re-deriving the live responses from scratch. It deliberately does not import models.forecasting or api.main - a shared helper would only prove the same code ran twice.

- Loads the two joblib artifacts and the feature parquet, then re-implements the recursive 1-step-feedback forecast and the calendar/lag/rolling features independently.
- Compares its own p50 curve to the live API's, step by step, for all three zones, and re-derives P10/P90 from forecast_intervals.json.
- Re-runs the risk classifier on the exact historical row /risk says it scored, to 12 decimals.
- Runs two probes a fabricated constant would fail: shift the input history by +5% and +10% and require the outputs to move.

largest p50 mismatch across zones : 0.000e+00 MW
checks run 33 | passed 33 | failed 0

33 checks passed, 0 failed. The live API's forecast and risk numbers are reproduced exactly by the committed LightGBM artifacts applied to the committed feature data - nothing served is a hard-coded value.

6. Model performance on the real data

1-step-ahead on the held-out test window (20% most recent, time-ordered split)

Model	MAPE	MAE (MW)	RMSE (MW)
Persistence (t-24h)	7.81%	842.0	1,640.5
Seasonal-naive (t-168h)	9.20%	788.7	1,249.4
LightGBM	2.10%	176.5	278.6
LightGBM no-weather	2.12%	180.0	284.4

LightGBM cuts 1-step MAPE from 9.20% (seasonal-naive) to 2.10%, a 77.1% relative improvement. The band is not a fixed percentage: it is split-conformal relative-error quantiles per horizon, calibrated on the validation window and evaluated once on the test window (82.8% empirical coverage of a nominal 80% band across 834 origins).

Replayed forecast error and band coverage by horizon

Horizon	LGBM MAPE	Naive MAPE	Improvement	P10-P90 coverage	Bias %
h+1	2.12%	9.27%	77.2%	78.8%	+0.17%
h+6	4.99%	9.26%	46.1%	84.1%	+0.75%
h+12	5.45%	9.26%	41.2%	86.1%	+0.90%
h+24	5.71%	9.27%	38.4%	83.9%	+1.04%

Release gates from the closed-loop replay

Method: closed-loop replay of the held-out test window through the serving path: recursive LightGBM forecast + split-conformal band + risk classifier + decision_rules alarm/shed thresholds, scored against realized load

Gate	Family	Measured	Target	Floor	Result
h+1 forecast beats seasonal-naive by >=10% MAPE	forecast	0.7717	0.1	0	PASS
h+24 forecast beats seasonal-naive by >=10% MAPE	forecast	0.3841	0.1	0	PASS
P10-P90 coverage within 0.70-0.95 (nominal 0.80)	forecast	0.8492	[0.7, 0.95]	[0.6, 0.98]	PASS
weekly coverage spread <=0.15 (band not decaying across the window)	forecast	0.1948	0.15	0.3	WARN
overload warning raised 6h ahead is >=50% correct (all origins, not just events)	forecast	0.7641	0.5	0.35	PASS
alarm precision >=0.50 at score>=0.6	alarm	0.8838	0.5	0.35	PASS
alarm recall >=0.50 at score>=0.6	alarm	0.7844	0.5	0.35	PASS
Brier skill over the base-rate forecast >=0.15	alarm	0.7204	0.15	0.05	PASS
<=5 false alarms per zone-day (operator alert fatigue)	alarm	0.313	5	10	PASS

Gate	Family	Measured	Target	Floor	Result
>=80% of real overload events raise an alarm while running	event	0.7845	0.8	0.6	WARN
>=50% of events forecast-warned >=3h before onset	event	0.3425	0.5	0.25	WARN
median STABLE forecast lead before onset >=3h (unbroken warning from h+1)	event	1	3	1	WARN
>=500 forecast origins replayed (window actually covered)	data	1.005e+04	500	100	PASS

WARN: 9/13 gates pass, 4 warn, 0 fail on 3,464 replayed test hours (10,122 rows scored, 10,050 forecast origins).

The one gate that went from PASS to FAIL, and why that is the honest result

median_lead_hours asks whether the p50 forecast puts the onset hour of an overload episode itself at or over the warning line, unbroken from h+1. On the real pull the median is 1 h and only 57% of events are warned even one hour ahead. This was measured, not assumed: 54% of all overload hours in the served table sit within five percent of the 0.9 x capacity line, and a forecast with ~2% MAPE cannot reliably clear a threshold the actual value only crosses by one to three percent. The synthetic grid had smooth deterministic ramps, so the same gate passed there - the failure is the real data telling us the point forecast is not an advance-warning device for marginal overloads.

The rule is not useless, and the gate set is not one-sided. Scored over all origins with a precision guard, the same warning rule holds 0.92 precision / 0.91 recall at h+1 and 0.71 / 0.75 at h+24. Widening to the upper band would buy a median nominal lead of 24 h but drops h+6 precision to 0.40 on 3,175 alerts - an almost-always-warn rule, which is exactly the failure mode a release gate suite exists to reject. The threshold was therefore left alone and the finding reported. What the operator is actually paged on - the risk-score alarm path - passes every gate in its family: precision 0.884, recall 0.784, 0.31 false alarms per zone-day, Brier skill +0.720.

What an operator would have experienced

Alarm precision	0.884 at threshold 0.6
Alarm recall	0.784 over 10,122 scored zone-hours
False alarms	0.31 per zone-day (132 false hours)
Brier skill	+0.7204 vs base rate (0.0309 vs 0.1105)
Overload events	142 of 181 detected (78%)
Warning lead	median 1 h stable / 5 h glimpse on the p50 rule; 34% warned >=3 h
Shed recommendations	994 priority-1 hours, 142 priority-2 hours
Peak onsets (UTC)	09Z x24, 08Z x23, 07Z x20, 10Z x20

Bucket	Hours	Mean predicted	Observed overload rate
low	8,813	0.007	0.022
moderate	173	0.452	0.468
elevated	142	0.705	0.613
critical	994	0.973	0.922

Predicted risk tracks observed overload strongly at the ends but is not strictly monotone in the middle: the moderate and elevated buckets swap order inside overlapping 95% binomial intervals (n=173 and 142 hours). The validator now appends that inversion to the shipped report's own limitations automatically, and the release test enforces ordering only where the sample is large enough to demand it.

7. Bugs found in the running system and fixed

Found by auditing the deployed service, not by reading the source. Two findings raised by review agents were discarded after measurement proved them wrong - a colour-alpha helper whose computed style was valid, and a missing abort guard that was already present - so this table is only what survived verification.

#	Defect	Evidence	Fix
1	/risk returned the observed load of a historical snapshot in a field named predicted_load_mw	Dashboard showed 6,117 MW 'predicted' beside a 6,905 MW step-1 forecast for the same zone	Renamed to snapshot_load_mw, scored hour surfaced in the panel, mocks updated
2	/forecast indexed lags by row position but stamped output from the final timestamp	An 83-row hole left an 80-row run that still satisfied the 168 h minimum on rows no longer corresponding to hours	_contiguous_hourly_tail() trims to the gap-free suffix; a 503 names the shortfall
3	Risk-ring colours vanished in the 3D topology	three.js r185 Color parses only legacy comma-separated hsl(); the space-separated CSS string silently	toThreeColor() converts the token back to hsl(h, s%, l%) before it reaches THREE.Color
4	/shed-schedule built a reasoning string claiming 'Predicted load' from snapshot values	The test asserted a forecast the endpoint never made	Reasoning now states measured load at the scored hour and labels capacity as a proxy
5	An unreachable guard in the shed-schedule path	The branch could never fire because /assets already filters the same condition	Removed, with the docstring claim it supported
6	top_factors could be None and crashed the detail panel	RiskDetail mapped over a possibly-null list	(risk.top_factors ?? []) plus Number.isFinite guards on load and capacity
7	Metric and version claims in README/docs had drifted from the artifacts	PR-AUC, coverage, endpoint count, test count, alarm burden, three.js and recharts versions all stale; one docs link pointed at a nonexistent file	Every figure re-read from its artifact; dead link replaced

8. Documentation corrections

Where	Stale claim	Now reads
README.md	PR-AUC 0.9637 / coverage 84.8% / 6 endpoints / 47 tests / 145 alarm-days	Re-read from artifacts; /risk sample replaced with the payload the running service returns
docs/03_System_Architecture.md	Diagram listed 5 of 7 routes	All 7 routes listed
docs/08_Testing_Validation_Prompt.md	Linked src/GridSenseDashboard.jsx, which does not exist	Links a verified GitHub file with env vars confirmed at the cited lines
docs/11_Tech_Stack.md	Three.js 0.128.0, Recharts 2.12.7	Versions read from package-lock.json
docs/13_Closed_Loop_Validation_Update.md	16 KB report, 47 tests, commit SHAs from the pre-merge review branch	Actual byte size, collected test count and current HEAD SHAs
ingestion/entsoe_client.py	Never exercised against the live API; comment claimed it worked	Six real ingestion defects fixed and documented from live responses

9. Deployment verification

Live endpoints

Route	HTTP
/health	200
/validation	200
/assets	200
/forecast/ES	200
/forecast/GR	200
/forecast/IT_SOUTH	200
/risk/ES	200
/risk/GR	200
/risk/IT_SOUTH	200
/shed-schedule/IT_SOUTH	200
/forecast/NOPE	404

/forecast/NOPE is a deliberate negative control: an unknown zone must not return 200.

Builds, tests, repositories, CI

Backend commit	fcf0744 working tree clean
Frontend commit	b44f979 UNPUSHED CHANGES PRESENT
pytest	59 collected - 59 passed, 1 warning in 35.99s
Backend CI	ci: success; ci: success; ci: success
Frontend CI	ci: success; ci: success; ci: success
Site	https://grid-senseai.vercel.app responded HTTP 200
Serving window	2026-10-21T18:00:00+00:00 to 2026-11-04T17:00:00+00:00
Interval method	split-conformal relative-error quantiles of the recursive weather-aware LightGBM forecaster, calibrated on the validation window, evaluated once on the test window

GET /health reports 'ok' with all 6 artifacts loaded and declares is_synthetic=False.

GET /validation on the live host serves the same artifact committed in the repo: overall WARN, 13 gates, window 2026-06-13 onward.

10. Limitations, stated plainly

The first block is emitted by models/closed_loop_validation.py itself - including an auto-appended line for every gate that fails and every inversion in the reliability ladder, so the shipped artifact cannot read as green by omission. The second block is the standing product limitations.

- Future weather is persistence (same hour on the most recent observed day), not a real NWP feed, so error during weather events is understated.
- capacity_mw is a p99-of-load proxy: 'overload' means unusually high for this zone, not a verified thermal emergency. Absolute risk values inherit that.
- The live /risk endpoint scores one historical snapshot matched to the current weekday+hour; this replay scores every hour in sequence, so it measures the model, not the snapshot-selection rule.
- Labels derive from the same load series the models consume, so a sensor outage or curtailment that suppressed measured load would read as 'no overload'.
- Overload onsets cluster at each zone's daily peak hour, so a long 'lead time' can come from reproducing the climatological peak shape rather than new information. Read it with the h+6 warning-rule precision gate, which scores the rule over all origins - including the hours where it cries wolf.
- The newest hour this replay can score is 26 h behind wall clock. Load arrives from ENTSO-E as it is republished, and NASA POWER only exposes an hour roughly two days after it happens; a row missing any weather input is dropped rather than fed to the model with NaNs, because LightGBM silently predicts through missing features. So the served origin is the last *fully observed* hour, not the current one. A near-real-time provider was measured as a substitute and rejected for now: it tracks NASA on temperature (r=0.92, bias +0.4 degC) but not on wind (1.8 vs 2.9 m/s, r=0.62), and splicing it into the tail would score the forecast against a feature distribution the model was never trained on.
- No refresh path. api/main.py loads the committed feature frame at startup and keeps the newest rows per zone, and makes no outbound request at all, so the served window ages one hour per hour after each deploy - measured: newest hour 2026-11-04T17:00 against a build time of 2026-11-06 07:49 UTC. The ages in /health and in the UI report this truthfully, so it is a missing pipeline stage and not a mislabel. Two properties come with it: the runtime holds no third-party credentials, and it cannot be failed or attacked through an upstream fetch at request time.
- capacity_mw is a per-zone p99 of observed load, so 'overload' means unusually high for that zone - not a verified thermal limit. Every risk value inherits that, including the 0.6 alarm and 0.8 shed cuts.
- That p99 is computed over the whole series, so the definition of the label sees the test window. No model is trained on test rows, but the label is mildly transductive and the absolute rates are therefore optimistic in a way a clean-room holdout would not be.
- Weather is a reanalysis product (NASA POWER), roughly two days behind real time: the system can score the recent past honestly but cannot see tomorrow's weather, which is why future weather is persistence.
- The usable window ends 2026-11-04 17:00 UTC, which is where both sources exist - not a forecast horizon. A dashboard reading on a later date is reading a stale anchor.
- Three European zones. The stated target domain is Pakistan's NTDC/DISCO grid; nothing here has been run on that data and nothing here should be read as claiming it transfers unchanged.
- The live /risk score is one historical snapshot matched to the current weekday and hour - a replay, not a nowcast, which is why the panel shows the scored hour.

- No load-duration curve, outage log or shed history was available, so the counterfactual 'what would this have saved' is deliberately absent from every claim in this report.

11. Committed work and future direction

Ordered by what actually remains. 'Status' is the honest current state at build time, not a plan.

Commitment	Why it matters	Acceptance criterion	Status
Push the real-data artifacts and redeploy	The repo and the live service must not disagree: /health on the deployed host still reports the bootstrap until the new parquet and joblib files land	Live /health declares load=entsoe, weather=nasa_power, is_synthetic=false; integrity checker passes against the live host	OPEN
Advance warning for marginal overloads	54% of overload hours sit within 5% of the line and the p50 rule cannot warn on them; this is the failing gate	A named, precision-guarded rule achieves a documented lead on marginal events, or the gate is replaced by an explicitly re-scoped operator promise	OPEN
Recalibrate the middle risk bands	moderate/elevated observed rates invert inside overlapping CIs, so the dashboard's middle bands are coarser than their labels suggest	Isotonic or Platt recalibration on the validation split, monotone reliability with CIs reported per bucket, SHAP bucket explanations regenerated from the same	OPEN
Live refresh instead of a baked parquet	The file-based serving path is a pilot shortcut: data behind real time makes every 'current' number stale by up to two days	Scheduled ingestion -> retrain -> artifact swap -> smoke test, with serving_window_end within the source's own publication lag	OPEN
Numerical weather input (NWP) feed	Persistence weather understates error during exactly the events that matter - a heat wave or a wind ramp	Replace persistence with Open-Meteo/ICON forecast inputs, then re-calibrate the conformal bands on forecast weather rather than reanalysis	OPEN
Rated capacity and outage history	Removes the self-referential p99 label and turns risk into a physical statement	Per-zone rated MW from the TSO or an official shed/outage log; labels rebuilt from it; the proxy flag retired	OPEN
Run the pipeline on NTDC/DISCO data	The pitch targets Pakistan; the evidence is Europe	Same chain executed on Pakistani zones with published MAPE and gate table for those zones, or the claim is removed from the	OPEN
Integrity checker as a CI job	'The served number equals the model output' should stay a checked fact, not a report claim	Scheduled workflow runs the checker against the deployed host and fails on any [FAIL] line	OPEN
Test suite and gate policy	The release policy was tightened while migrating: an adverse gate may ship only if the report explains it	pytest green; every FAIL named in limitations; gap handling covered by the three new regression tests	DONE in this build - see the measured suite line above

Reproduce every figure in this document

```
python scripts/build_audit_report_pdf.py
python scripts/verify_prediction_integrity.py
python -m models.closed_loop_validation --strict
python -m pytest -q
```

This generator is committed in the backend repo. It reads only committed artifacts and live endpoints; if a figure below the fold disagrees with data/models/closed_loop_report.json, the artifact is right and this PDF is stale - rebuild it.